



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B.Tech (Biotechnology) (CBCS) Degree Examinations

December 2023

MA2002-1 – NUMERICAL METHODS

(For BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- The polynomial which represents the following data: $f(0)=5$ and $f(2)=1$
 A) $5-2x$ B) $2x+6$
 C) $6x+1$ D) $5x-1$
- The 4th order forward difference of the polynomial $5x^3 + 4x + 15$ is _____
 A) $15h^2$ B) A polynomial of degree 2
 C) 1 D) 0
- Newton- Gregory Forward difference interpolation formula can be used _____
 A) only for equally spaced intervals B) only for unequally spaced intervals
 C) for both equally and unequally spaced intervals D) All the above
- In case of Newton's backward difference formula which equation is correct to find p _____
 A) $x - x_0 = ph$ B) $x + x_0 = ph$
 C) $x - x_n = ph$ D) $x + x_n = ph$
- The value of $\int_0^1 \frac{1}{1+x} dx =$ _____ calculated by using Trapezoidal rule with $h=0.5$
 A) 0.6970 B) 0.6931
 C) 0.7084 D) 0.8945
- Using backward differences, the formula for $f''(x) =$ _____
 A) $\frac{1}{h} \left\{ \nabla y_n + \frac{(2p-1)}{2} \nabla^2 y_n + \frac{(3p^2-6p+2)}{6} \nabla^3 y_n + \dots \right\}$ B) $\frac{1}{h} \left\{ \nabla y_n + \frac{(2p+1)}{2} \nabla^2 y_n + \frac{(3p^2+6p+2)}{6} \nabla^3 y_n + \dots \right\}$
 C) $\frac{1}{h^2} \left\{ \nabla^2 y_n + (p-1) \nabla^3 y_n + \frac{(6p^2-18p+11)}{12} \nabla^4 y_n + \dots \right\}$ D) $\frac{1}{h^2} \left\{ \nabla^2 y_n + (p+1) \nabla^3 y_n + \frac{(6p^2+18p+11)}{12} \nabla^4 y_n + \dots \right\}$
- Given

x	0	1	2	3
$y(x)$	2	4	8	14

 then $y'(0) =$ _____
 A) 2 B) 1
 C) 0.5 D) 0
- Given $(n+1)$ set of tabular values (t, a) of time and acceleration, then we can obtain the velocity of the particle by process of _____
 A) numerical integration B) numerical differentiation
 C) interpolation D) extrapolation
- Using modified Euler's method for the initial value problem $\frac{dy}{dx} = -xy^2$, $y(0) = 2$, $h = 0.1$ and the initial approximation $y_1^{(0)} = 2$, the value of $y_1^{(1)}$ is _____
 A) 2 B) 1.98
 C) 0.98 D) 0.99

10. For the initial value problem $\frac{dy}{dx} = f(x, y)$; $y(x_0) = y_0$, the Taylor's series expansion up to 4th order is
- A) $y_0 + h^2 y_0' + \frac{h^3}{2!} y_0'' + \frac{h^4}{3!} y_0''' + \frac{h^5}{4!} y_0^{iv} + \dots$ B) $y_0 + h y_0' + \frac{h^2}{2!} y_0'' + \frac{h^3}{3!} y_0''' + \frac{h^4}{4!} y_0^{iv} + \dots$
- C) $y_0 + h y_0' - \frac{h^2}{2!} y_0'' + \frac{h^3}{3!} y_0''' - \frac{h^4}{4!} y_0^{iv} + \dots$ D) $h y_0' + \frac{h^2}{2!} y_0'' + \frac{h^3}{3!} y_0''' + \frac{h^4}{4!} y_0^{iv} + \dots$
11. Milne's corrector formula to find y_4 is _____
- A) $y_4 = y_2 - \frac{h}{3}(f_2 + 4f_3 + f_4^{(P)})$ B) $y_4 = y_2 + \frac{h}{3}(f_2 + 4f_3 - f_4^{(P)})$
- C) $y_4 = y_2 + \frac{h}{3}(f_2 - 4f_3 + f_4^{(P)})$ D) $y_4 = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4^{(P)})$
12. In Runge-Kutta method of order four, the value of k_1 for the initial value problem $\frac{dy}{dx} = 1 + y^2$, $y(0) = 0$, $h = 0.2$ is _____
- A) 0.2 B) 0.22
- C) 0.1 D) 0.15
13. Which of the following is one-dimensional wave equation?
- A) $u_x = c^2 u_{xx}$ B) $u_t = c^2 u_{xx}$
- C) $u_{tt} = c^2 u_{xx}$ D) $u_{xx} = c^2 u_{tt}$
14. For the wave equation $u_{tt} = u_{xx}$, $h = k = 0.2$, the value of α is _____
- A) 1 B) 1.5
- C) 0.5 D) 1.25
15. The Schmidt explicit formula for finding the solution of heat equation $u_t = c^2 u_{xx}$ is _____
- A) $u_{i,j+1} = \alpha [u_{i-1,j} + u_{i+1,j}] + 2\alpha u_{i,j}$ B) $u_{i,j+1} = \frac{1}{2} [u_{i-1,j} + u_{i+1,j}]$
- C) $u_{i,j+1} = \alpha [u_{i-1,j} - u_{i+1,j}] + (1 - 2\alpha)u_{i,j}$ D) $u_{i,j+1} = \alpha [u_{i-1,j} + u_{i+1,j}] + (1 - 2\alpha)u_{i,j}$
16. For the heat equation $u_t = \frac{1}{4} u_{xx}$, $h = k = 1$, the value of mesh ratio parameter α is _____
- A) 1 B) 0.5
- C) 0.25 D) 2
17. The root of the equation $xe^x = \cos x$ lies in the interval _____
- A) (0, 1) B) (1, 2)
- C) (2, 3) D) (-1, 0)
18. Newton Raphson method can be used to solve
- A) a system of non-linear equations B) a system of linear equations
- C) both linear and non-linear equations D) none of these
19. The first approximation to the root of $x^3 - 2x - 5 = 0$ using bisection method is _____
- A) 1.5 B) 1.55
- C) 2.5 D) 2.55
20. Gauss Seidel method is also termed as
- A) Elimination method B) Successive iteration
- C) False position D) none of these

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

1. a) Derive Newton's forward difference interpolation formula.
b) Find the missing term in the following table:

x	0	1	2	3	4
y	1	2	-	82	257

- c) Using Simpson's $\frac{3}{8}$ th rule, evaluate $\int_0^{0.3} \sqrt{1-8x^3} dx$ by taking 7 ordinates.

Marks	BT*	CO*	PO*
5	L*2	1	1
5	L3	1	2
6	L3	2	2

2. a) The velocity 'v' of a particle at distance 's' from a point on its path is given by the table below:

s in meters	0	10	20	30	40	50	60
v in m/sec	47	58	64	65	61	52	38

Estimate the time taken to travel 60m by using Simpson's $\frac{1}{3}$ rd rule.

5 L4 2 1

- b) Evaluate $\int_0^{\pi/2} \sqrt{\cos \theta} d\theta$ by dividing the interval into 6 equal parts using

Weddle's Rule.

5 L2 2 2

- c) Given that:

x	5	7	11	13	17
y	150	392	1452	2366	5202

Evaluate f(9) using Newton's divided difference interpolation formula.

6 L3 1 2

3. a) Using Lagrange's Interpolation formula find f(4) given that f(0)=0, f(2)=4, f(3)=27, f(5)=175.

5 L3 1 1

- b) Derive Newton's general quadrature formula and deduce Trapezoidal rule.

5 L2 2 2

- c) Given:

x	30	40	50	60	70
f(x)	2.613	6.244	12.255	21.246	33.817

Find first and second derivatives of the function at x = 30.

6 L3 2 2

Unit – II

4. a) Using fourth order R-K method, find y(1.1) given that $y' = 3x + y^2$ and y(1)=1.2. Take h = 0.1.

5 L3 3 1

- b) Given that $\frac{dy}{dx} = xy + y^2$, y(0) = 1 ; y(0.1) = 1.1169 ; y(0.2) = 1.2773 ; y(0.3) = 1.5049. Compute y(0.4) by Milne's Predictor – Corrector method.

5 L3 3 1

- c) Solve the parabolic equation $u_{xx} = 2u_t$ subjected to the condition $u(0, t) = 0 = u(4, t)$; $t \geq 0$ & $u(x, 0) = x(4 - x)$. Take h = 0.1. Carry out computation for 4 levels.

6 L4 4 2

5. a) Using Modified Euler's method find the approximate value of y at x=0.2 given that $y' = x + y$ and y(0) = 1. Take h = 0.1. Carry out 2 iterations in each stage.

5 L2 4 2

- b) Solve the PDE $\nabla^2 u = -10(x^2 + y^2 + 10)$ over the square with sides $x = 0 = y$; $x = 3 = y$ with $u = 0$ on the boundary & mesh length equal to one.

5 L3 4 2

- c) Solve the wave equation in the interval $0 \leq x \leq 1$; $t \geq 0$; $u(x, 0) = 100 \sin \pi x$; $\frac{\partial u}{\partial t} = 0$; $u(0, t) = 0 = u(1, t)$.

Take $h = \frac{1}{4}$. Compute u for 4 steps.

6 L3 3 1

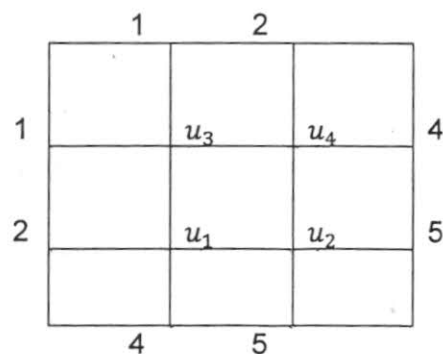
6. a) Given that $\frac{dy}{dx} = x^2(1 + y)$; y(1) = 1 ; y(1.1) = 1.233 ; y(1.2) = 1.548 ; y(1.3) = 1.979. Evaluate y(1.4) using Adam's Bashforth Predictor – Corrector formula. Apply corrector formula twice.

5 L1 3 1

- b) Employ Taylor's series method to compute y(0.2) given that: $y' = 1 - 2xy$ and y(0) = 0.

5 L2 3 1

- c) Solve the Laplace equation $u_{xx} + u_{yy} = 0$ at the interior nodes of the following square mesh (given below) taking the values provided at the boundary using standard 5 point formula.



Unit – III

7. a) Using the Bisection method obtain the root of the equation $x^3 - 4x - 9 = 0$. Carry out 5 iterations.
- b) Using Newton Raphson method obtain the root of the equation: $\tan x = 1.5x$ by taking $x_0 = 1$. Carry out 4 iterations.
- c) Apply Gauss –Seidel iteration method to solve the system of equations given below. Carry out three iterations.

$$20x + y - 2z = 17$$

$$3x + 20y - z = -18$$

$$2x - 3y + 20z = 25$$

$$\text{Take } x^0 = y^0 = z^0 = 0$$

8. a) Find the root of the equation $x^3 - 2x - 5 = 0$ in (2,3) using the method of false position. Calculate the root correct to four decimal places. Carry out 3 iterations.
- b) Use the method of ordinary iteration to obtain the root of $x^4 - 16x + 16 = 0$ near $x = 1$. Carry out five iterations.
- c) Using Newton-Raphson method solve the system of non-linear equations. Carry out two iterations.

$$x^2 + y = 11 ; y^2 + x = 7 \text{ given } x_0 = 3.5, y_0 = -1.8$$

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
